

Learning rate

 $1 \times 10^{-4}$  $3 \times 10^{-4}$  $5 \times 10^{-4}$  $r=4, \alpha=4$  $r=4, \alpha=8$  $r=8, \alpha=8$  $r=8, \alpha=16$  $r=16, \alpha=16$  $r=16, \alpha=32$  $r=32, \alpha=32$  $r=32, \alpha=64$  $r=64, \alpha=64$  $r=64, \alpha=128$  $(r, \alpha)$ 

0.957

0.961

0.960

0.962

0.962

0.962

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0.961

0.956

0.959

0.960

0.957

0.956

0.961

0.956

0.638

0.955

0.818

0.640

0.011

0.95

0.90

0.85

0.80

Mean f1